

Kothe Viswanath

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Technical Skills

Programming & Scripting: Python, SQL, Bash, Go

DeepLearning & ML: PyTorch, TensorFlow, Hugging Face Transformers, QLoRA, LoRA, PyTorch Lightning, XGBoost

LLM & AI Tools: LangChain, LlamaIndex, LangGraph, DeepAgents, MCP, RAG pipelines, Multi-agent workflows

Data Deployment: Spark, Flink, Kafka, Airflow, Docker, Kubernetes, SLURM, KubeFlow, AWS (S3), (Feast), CI/CD

Monitoring & Evaluation: MLflow, Prometheus, Grafana, OpenTelemetry, Data validation, Anomaly detection, BLEU, ROUGE, nDCG, Top-K accuracy

Experience

AI/ML Research Assistant

Syracuse University

Jun 2025 – Present

- Supported development of an **Agentic AI-based autofuzzer**, improving test code coverage by **~15%**.
- Fine-tuned LLMs for code generation**, reducing hallucinations by **40%** and achieving **90%+ compile success rate** across generated samples.
- Built **large-scale data pipelines** (Airflow + Kafka + Python) processing **5M+ code files** into a **RAG knowledge base** with lineage tracking.
- Designed and evaluated **RAG pipelines** using BLEU, ROUGE, recall@k improving retrieval accuracy by **~28%**.
- Collaborated cross-functionally to define requirements, implement **guardrails**, and designed automated evaluation suites, improving reliability.

Machine Learning Engineer

Indian Space Research Organisation (ISRO), Bangalore, India

Jan 2022 – Jun 2023

- Led development of an **end-to-end AI/ML monitoring system** reducing anomaly resolution time by **35%** and lowering operator workload.
- Deployed deep learning (LSTM and Transformers) models on **multi-GPU clusters** using PyTorch DDP/SLURM, achieving **4.2x training speedup** and **lower inference latency**.
- Built and maintained **telemetry ingestion + validation pipelines** (Kafka + Spark), handling **20k events/sec** with less than 200ms latency.
- Implemented full **CI/CD + observability** using Kubernetes, ArgoCD, GitHub Actions, OpenTelemetry, MLFlow, and Grafana; established **auto-retraining workflows** triggered by accuracy drift.
- Coordinated with scientists, compliance teams, and domain experts to ensure safety, accuracy, and mission reliability.

Data Scientist

Nuviso Networks, Bangalore, India

Jun 2021 – Dec 2021

- Trained predictive models** for **forecasting** and **anomaly detection** models (XGBoost, SARIMAX, Prophet), improving early detection and reducing false positives by 10-12%.
- Performed **exploratory data analysis** and **feature discovery** for datasets up to **50M+ rows**, improving model performance by **~15%**.
- Built scalable **ETL pipelines** (Python + SQL + Airflow) reducing data processing time by **30%**.
- Supported deployment by containerizing services with **Docker**, adding automated tests, and integrating builds into CI/CD.

Personal Projects

JarVish: <https://github.com/vishwa419/JarVish>

Oct 2025 – Present

Jarvis clone, powered by **GPT, deepagents, subagents, RAG** and **MCPs** as **microservice**. Powering end to end daily workflow automation with observability (LangChain, LangGraph, Qdrant, Docker, Opik).

AGENTIC DOCKER: https://github.com/vishwa419/Kitty_containers

Aug 2025 – Sep 2025

Built a modular **agentic-AI system** for task orchestration, integrating GPT, MCP and RAG to automate container deployments (Python, LangGraph, LangChain, Qdrant).

MLOPS for recommendation: https://github.com/vishwa419/MLOps_Inference

Jun 2025 –

Aug 2025

GitOps-driven ML pipeline for automated data validation, feature store, model training, deployment, and monitoring across environments (Python, BentoML, MLFlow, Docker, Grafana, Great Expectations, Feast).

Novel Algorithm for explainable CNNs: <https://doi.org/10.1109/ICCCNT56998.2023.10306989>

Education

